Abstract

Rapid urbanisation and motorisation in emerging cities have intensified challenges related to traffic congestion and air pollution. This study uses Gurugram, Haryana (India) as a representative case of a rapidly urbanising Indian city to examine trends in air quality, traffic growth, and private vehicle dependence. The research is based entirely on secondary datasets obtained from national (Government of India) and international sources, covering air pollution data for 2024 and national transport statistics for 2001–2020.

National transport statistics show that two-wheelers and cars grew at a compound annual growth rate (CAGR) of approximately 10% between 2010 and 2020, while buses grew at a comparatively low rate of around 4%, indicating a deepening structural dependence on private vehicles over public transportation. Over the same period, PM_{10} in Gurugram averaged 186 µg/m³ in 2024—nearly four times the WHO annual guideline of 45 µg/m³—with peak episodes exceeding 893 µg/m³ during winter months.

The study emphasises strategies for effective traffic management, the environmental benefits of reducing vehicular emissions, and the critical role of expanding public transportation systems. The primary contribution of this study lies in integrating city-level air quality analysis with national transport growth trends to highlight policy-relevant linkages for rapidly urbanising Indian cities. The analytical framework adopted here is replicable across similar urban regions, and the findings offer actionable, evidence-based recommendations for sustainable urban and transportation planning.

Keywords: Environmental health, pollution, PM10, public transport, public health, urban planning, congestion pricing.

Introduction

Rapid urbanisation and population growth have been accompanied by a proportional rise in environmental pollution, including air, water, and noise pollution (World Bank, 2023). Among these, air pollution has emerged as one of the most critical global environmental and public health challenges (WHO, 2023a, 2023b). Across many rapidly developing cities worldwide, Air Quality Index (AQI) levels have shown a persistent upward trend, reflecting increasing pressures from urban growth, motorisation, and inadequate transport planning. India is among the most affected countries, where a substantial proportion of cities experience chronic PM2.5 pollution and steadily deteriorating air quality conditions (The Times of India, 2026).

This study examines the city of Gurugram, located within the Delhi National Capital Region (NCR), as a representative case of a rapidly urbanising Indian city facing severe air quality challenges. Gurugram has consistently recorded “very poor” to “severe” AQI levels, as reported by the Central Pollution Control Board (The Times of India, 2025). These conditions highlight not only the intensity of air pollution in the region but also reflect broader challenges related to urban management practices, transport planning, and public behaviour affecting environmental sustainability and public health.

Alongside rapid urban expansion, Gurugram has experienced a substantial increase in private vehicle ownership, indicating a declining reliance on public transportation systems (MoRTH, 2023). The extensive use of private vehicles, often even for short-distance travel, reflects inefficiencies in multimodal transport infrastructure and limited public transport attractiveness. This pattern has contributed to increasing traffic volumes and the formation of congestion-prone zones, particularly around major commercial and business centres such as Cyber Hub. Persistent congestion in these

areas leads to elevated levels of air and noise pollution, increased travel delays, and significant time losses that adversely affect daily commuters, working professionals, and overall urban mobility.

Improving the quality of life in a city such as Gurugram requires strategic interventions in urban planning, transport governance, and traffic management. An efficient, accessible, and user-oriented public transportation system is widely recognised as a critical component of sustainable urban development and national economic growth. Enhanced public transport connectivity and service quality can encourage a modal shift away from private vehicle use, thereby reducing traffic congestion, noise pollution, excessive travel times, and associated environmental impacts.

The combination of ineffective traffic management, high private vehicle dependence, and deteriorating air quality has intensified environmental and public health concerns, indicating systemic shortcomings in urban planning, administrative coordination, and public awareness. Prolonged exposure to poor air quality has been linked to respiratory and cardiovascular diseases, including asthma, bronchitis, chronic obstructive pulmonary disease, lung cancer, and stroke, with adverse effects extending beyond human populations to urban ecosystems as a whole (CDC, 2023; WHO, 2023b; World Bank, 2023).

Within this context, the present study aims to provide data-driven insights to support urban re-planning, traffic management, and congestion control strategies. It is important to note that while air quality data are specific to Gurugram, vehicle registration and road infrastructure data are drawn from state-level (Haryana) and national (India) sources, given the absence of publicly available Gurugram-specific transport statistics. These broader datasets are used to contextualise local trends within a wider regional and national pattern of motorisation. The study addresses three specific objectives: (1) to characterise air quality trends in Gurugram during 2024, including seasonal and meteorological influences on pollutant concentrations; (2) to analyse patterns of vehicle growth and road infrastructure expansion in Haryana and India between 2001 and 2020; and (3) to synthesise these findings into policy-relevant recommendations for traffic management, public transport expansion, and emission reduction in rapidly urbanising Indian cities.

Literature Review

Previous studies have highlighted that electrification of the vehicle fleet alone is insufficient to eliminate urban air pollution, largely due to the growing contribution of non-exhaust emissions, particularly tyre wear particles (TWPs). Non-exhaust emissions, which include tyre wear, brake wear, and resuspended road dust, have emerged as dominant contributors to particulate matter (PM) pollution, especially $PM_{2.5}$ and PM_{10}. Recent studies estimate that TWPs alone account for approximately 13% of total $PM_{2.5}$ concentrations, making them a significant yet comparatively underregulated source of urban air pollution (Kang et al., 2025; Tian et al., 2024).

Experimental and real-world investigations further indicate that particulate emissions generated through tyre wear can substantially exceed regulated exhaust emission limits, underscoring the limitations of policies focused solely on tailpipe emissions. Non-exhaust sources are estimated to contribute nearly 60% of $PM_{2.5}$ and over 70% of PM_{10} in urban environments, surpassing the relative contribution of traditional exhaust pollutants such as carbon monoxide, nitrogen oxides, sulphur dioxide, and volatile organic compounds (Kang et al., 2025; Tian et al., 2024; Emissions Analytics, 2023). Due to their fine size, these particles are readily inhaled, posing a considerable public health burden.

At a global scale, TWPs are released in substantial quantities, with annual emissions estimated at approximately 1.3 million tonnes in Europe and several million tonnes worldwide. These particles

have also been identified as a major source of microplastic pollution. TWPs consist of synthetic polymers, heavy metals, and chemical additives, including polycyclic aromatic hydrocarbons and 6PPD, which have been associated with adverse toxicological effects. Their small size enables long-range atmospheric transport, resulting in continuous exposure in urban environments, where individuals may inhale thousands of tyre-related particles daily (Kang et al., 2025; Tian et al., 2024).

Beyond human health impacts, TWPs contribute to environmental degradation across terrestrial and aquatic ecosystems. In soils, they can reduce permeability and fertility, while in agricultural systems they interfere with root function and nutrient uptake. In aquatic environments, TWPs accumulate in rivers, lakes, and marine systems, entering food chains and posing risks to ecosystem integrity and food safety. These findings highlight the broader ecological implications of tyre wear pollution beyond urban air quality alone (Kang et al., 2025).

Several city-level case studies, including those from Singapore, Stockholm, and London, demonstrate that demand-management strategies such as congestion pricing and the expansion of public transportation systems can effectively reduce vehicular emissions and improve air quality. These studies emphasise the importance of integrated transport planning and policy interventions in addressing both exhaust and non-exhaust pollution sources (SFCTA, 2020; FHWA, 2019).

Despite increasing adoption of electric vehicles (EVs), existing research indicates that EVs are often heavier than conventional vehicles, resulting in higher tyre wear and associated particulate emissions. As a result, EVs may inadvertently exacerbate non-exhaust PM pollution if not accompanied by complementary measures such as improved tyre design, lightweight vehicle engineering, intelligent traffic management, and regulatory frameworks targeting non-exhaust emissions (Kang et al., 2025).

Although extensive research has examined air pollution, transport policies, and health impacts at global and regional scales, there remains a lack of city-specific, longitudinal analyses integrating traffic growth, private vehicle ownership, and air quality trends for rapidly urbanising Indian cities. In particular, limited attention has been given to Gurugram—one of India's fastest-growing urban centres and a prominent economic hub within the Delhi NCR—highlighting a research gap that this study seeks to address through a data-driven assessment of air quality, traffic growth patterns, and urban policy implications. Although the present study does not directly measure TWP concentrations, the dominant role of PM_{10} in Gurugram's pollution profile and the rapid growth in private vehicle ownership underline the relevance of non-exhaust emission research to the city's environmental management agenda.

Methodology

Data Sources

This study is based on secondary data collected from multiple publicly available and authoritative sources. City-level air pollution and meteorological data for Gurugram for the year 2024 were obtained from openly accessible datasets hosted on Kaggle (Tripathi, 2024), which compile and redistribute data originally sourced from official monitoring agencies. The year 2024 was selected as the most recent complete annual dataset available at the time of this study. National-level transport statistics covering the period 2001–2020 were collected from published reports of the Ministry of Road Transport and Highways (MoRTH), Government of India. The period 2001–2020 was selected to capture two decades of motorisation growth within the scope of publicly published MoRTH datasets. It should be noted that Gurugram-specific vehicle registration data were not available in disaggregated form; accordingly, state-level (Haryana) and national (India) data are used as proxies to contextualise local transport trends.

In addition, the study incorporates international case studies addressing urban air pollution and traffic management practices, along with supporting evidence from peer-reviewed literature and reports published by the World Health Organization and the World Bank on health impacts associated with air pollution. These sources were used to contextualize the findings from Gurugram within broader global and policy-oriented perspectives (MoRTH, 2023; WHO, 2023a, 2023b; World Bank, 2023; SFCTA, 2020; FHWA, 2019; Metz, 2018; Tripathi, 2024; Kang et al., 2025; Tian et al., 2024).

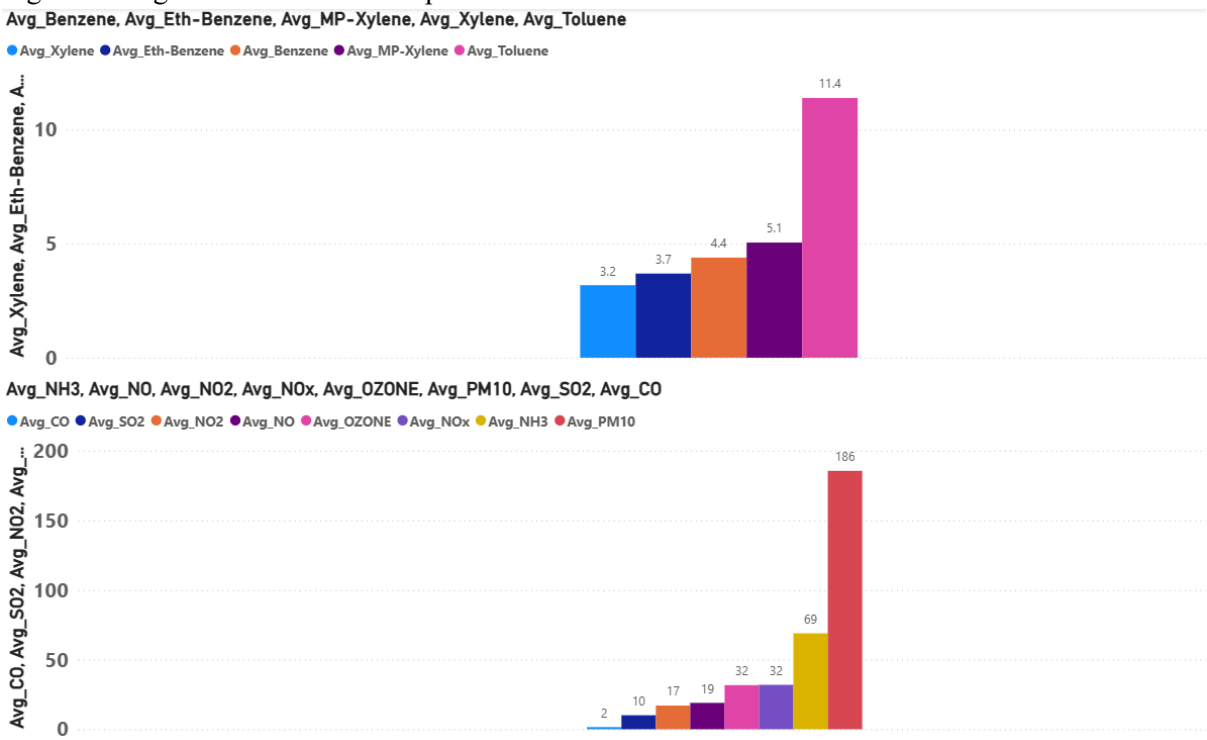
Data Processing and Analytical Tools

A combination of widely used analytical tools was employed to clean, process, analyse, and visualise the collected datasets. Microsoft Excel was used for initial data loading, basic preprocessing, and preliminary data cleaning to ensure consistency and usability. Tabula, an open-source data extraction tool, was utilised to extract tabular information from official government PDF reports and convert it into structured CSV formats.

Python was applied for further data refinement, including handling missing values and preparing datasets for analysis. MySQL, an open-source relational database management system, was used to execute structured queries and generate aggregated outputs required for analysis. Finally, Power BI was employed to develop interactive dashboards based on aggregated data tables, enabling the extraction and visual interpretation of key insights related to traffic growth, air pollution trends, and urban mobility patterns.

Results

Fig. 1. Average concentration of air pollutants in 2024.



According to this chart, the average concentrations of air pollutants in 2024 show distinct variation across different pollutant categories.

Table 1:

Xylene (µg/m³)	Eth-Benzene (µg/m³)	Benzene (µg/m³)	MP-Xylene (µg/m³)	Toluene (µg/m³)
3.2	3.7	4.4	5.1	11.4

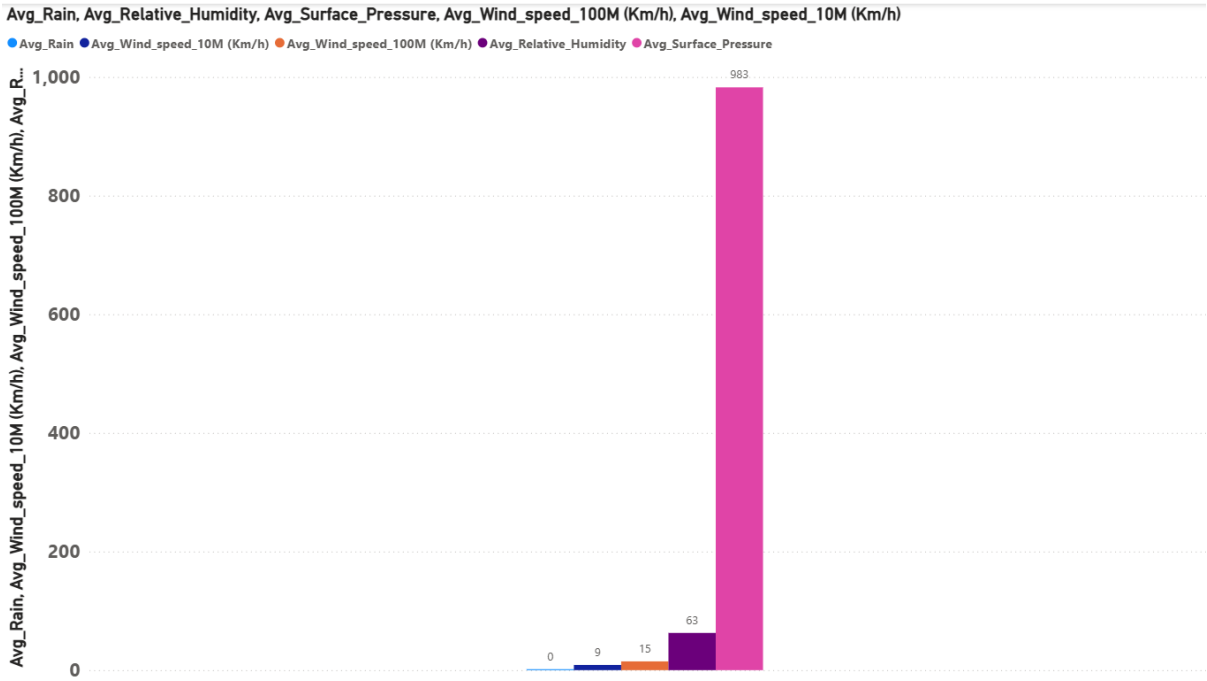
Table 1 shows that xylene recorded the lowest average concentration at 3.2 µg/m³, whereas toluene exhibited the highest concentration at 11.4 µg/m³.

Table 2:

CO (mg/m³)	SO2 (µg/m³)	NO2 (µg/m³)	NO (µg/m³)	Ozone (µg/m³)	Nox (ppb)	NH3 (µg/m³)	PM10 (µg/m³)
2	10	17	19	32	32	69	186

Table 2 shows that carbon monoxide (CO) recorded the lowest concentration at 2 mg/m³, while PM_{10} exhibited the highest concentration at 186 µg/m³, indicating a continuous increase in pollutant levels.

Fig. 2. Meteorological parameters influencing air quality in 2024.



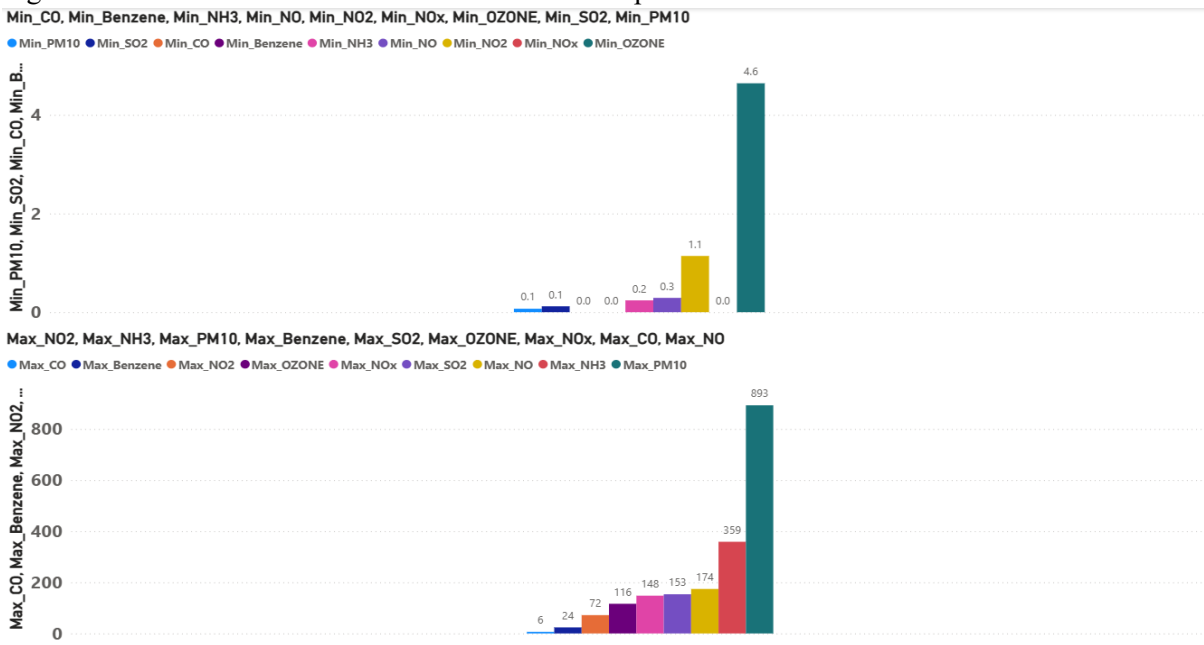
According to this chart, the average values of meteorological parameters in 2024 show clear variation across different factors.

Table 3:

Rain (mm)	Wind Speed_10M (Km/h)	Wind Speed_100M (Km/h)	Relative Humidity (%)	Surface Pressure (hPa)
0	9	15	63	983

Table 3 shows that rainfall recorded the lowest value at 0 mm, while relative humidity averaged 63%.

Fig. 3. Minimum and maximum concentrations of air pollutants in 2024.



According to this chart, the minimum and maximum concentrations of air pollutants recorded in 2024 show substantial variation across different pollutants.

Table 4. For Minimum

CO (mg/m³)	Benzene (µg/m³)	Nox (ppb)	PM10 (µg/m³)	SO2 (µg/m³)	NH3 (µg/m³)	NO (µg/m³)	NO2 (µg/m³)	Ozone (µg/m³)
0	0	0	0.1	0.12	0.2	0.3	1.1	4.6

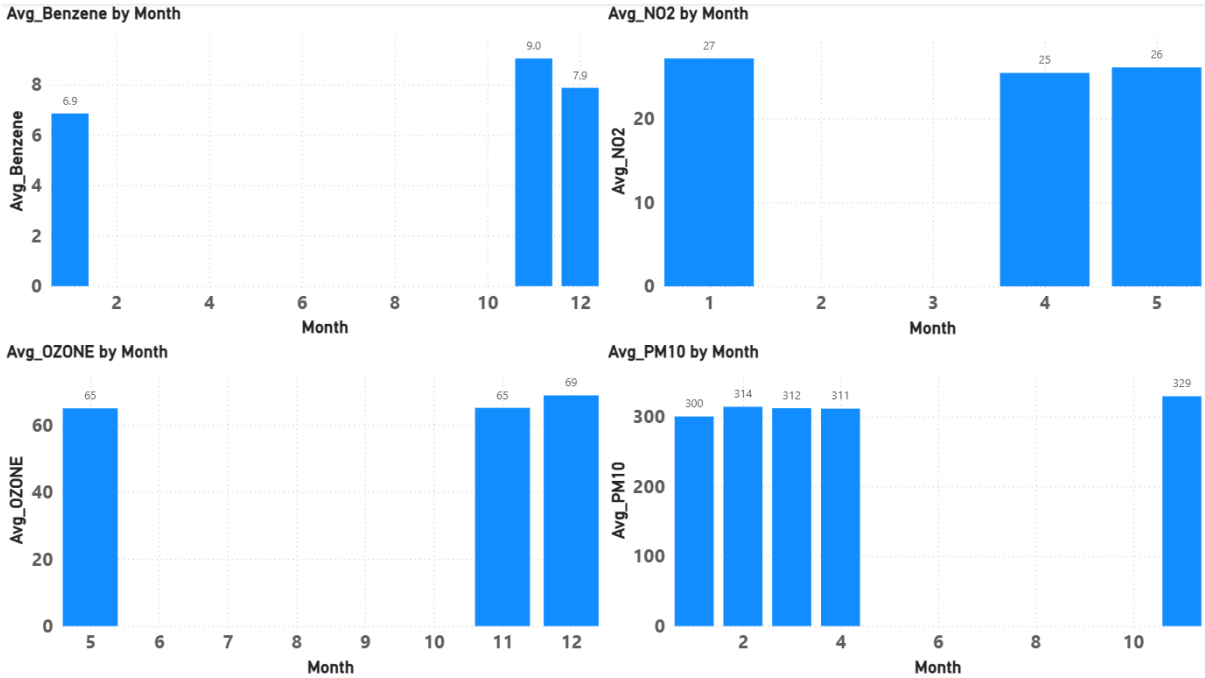
Table 4 shows that carbon monoxide (CO), benzene, and NO_x recorded zero values, while ozone exhibited the highest concentration at 4.6 µg/m³.

Table 5. For Maximum

CO (mg/m³)	Benzene (µg/m³)	NO2 (µg/m³)	Ozone (µg/m³)	Nox (ppb)	SO2 (µg/m³)	NO (µg/m³)	NH3 (µg/m³)	PM10 (µg/m³)
6	24	72	116	148	153	174	359	893

Table 5 shows that carbon monoxide (CO) recorded the lowest concentration at 6 mg/m³, while PM_{10} exhibited the highest concentration at 893 µg/m³, indicating a continuous increase in pollutant levels in the air.

Fig. 4. Monthly average concentrations of the top 10 highest levels of PM_{10}, NO_2, ozone, and benzene in 2024.



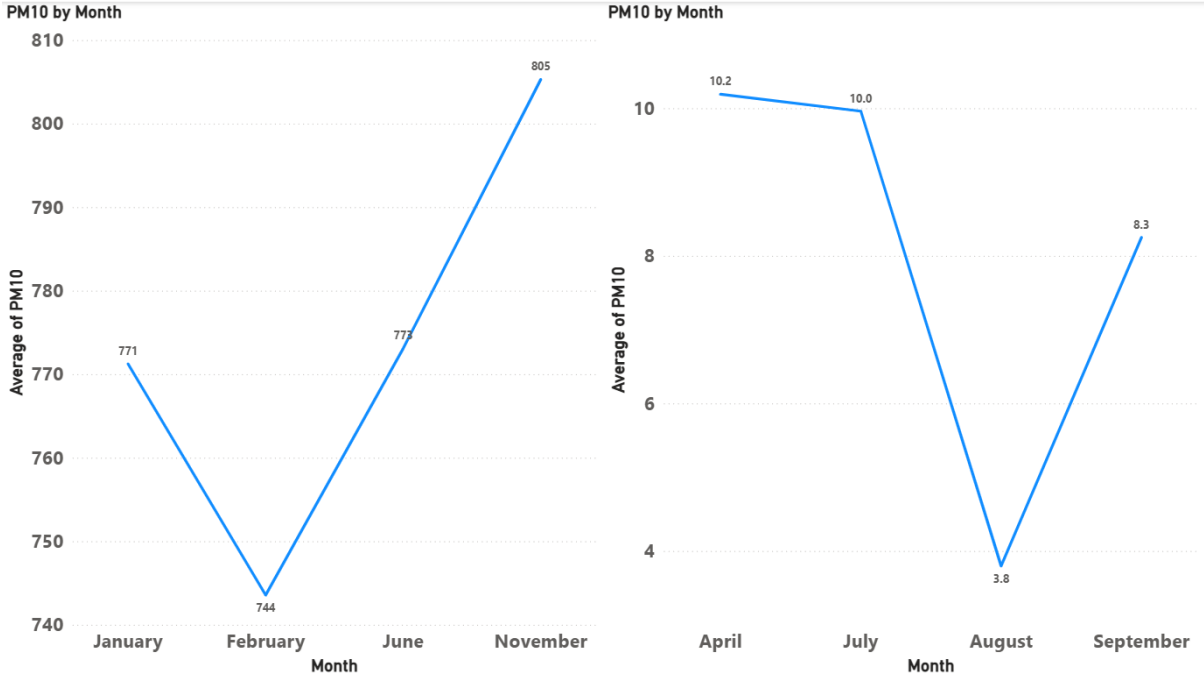
According to this chart, the monthly average concentrations of the top 10 highest values of selected pollutants in 2024 show distinct temporal variation.

Table 6:

	1	2	3	4	5	6	7	8	9	10	11	12
Benzene (µg/m³)	6.9										9	7.9
NO2 (µg/m³)	27			25	26							
Ozone (µg/m³)	65										65	69
PM10 (µg/m³)	300	314	312	311							329	

Table 6 shows that the beginning and end of the year recorded the highest pollution levels, with PM_{10} exceeding 299 µg/m³ in more months compared to other pollutants.

Fig. 5. Average of the top 10 highest and lowest PM_{10} concentration days in 2024.



According to this chart, the average PM_{10} concentrations during the top 10 highest and lowest pollution days in 2024 show a clear contrast.

Table 7: Minimum

	April	July	August	September
PM10 (µg/m³)	10.2	10	3.8	8.3

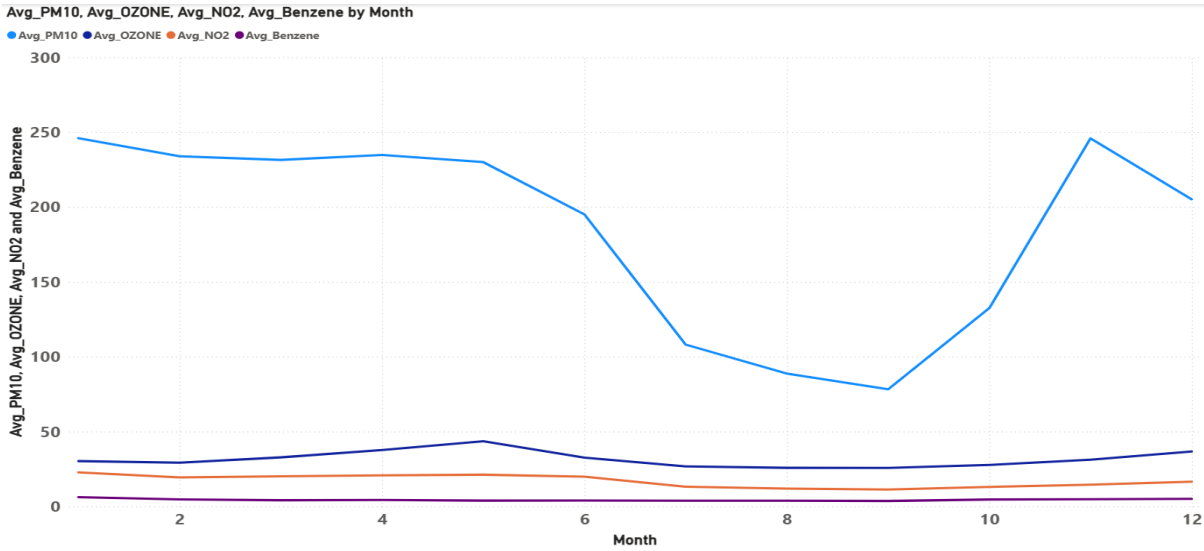
Table 7 shows that the lowest average pollution levels were observed during the middle of the year.

Table 8: Maximum

	January	February	June	November
PM10 (µg/m³)	771	744	773	805

Table 8 shows that the highest average pollution levels were predominantly observed during the first and last quarters of the year.

Fig. 6. Annual trend of average concentrations of the top four selected pollutants.



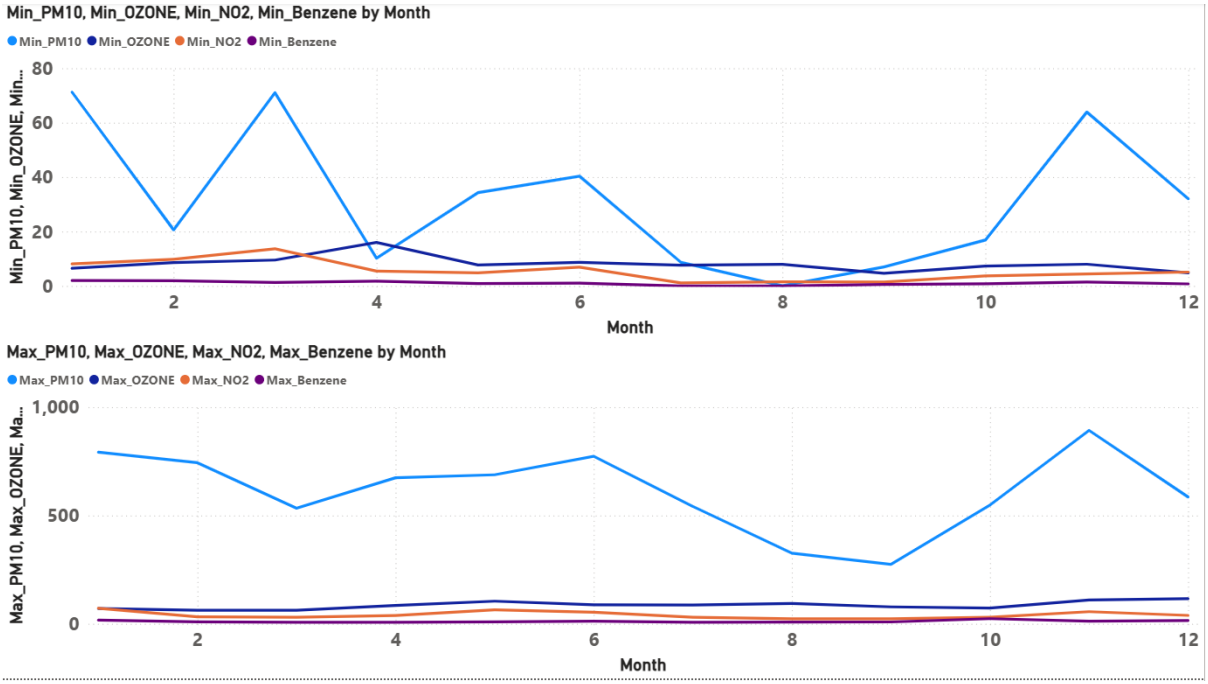
According to this chart, the annual trends of the average concentrations of four selected pollutants show clear temporal variation across the year.

Table 9:

	1	2	3	4	5	6	7	8	9	10	11	12
PM10 (µg/m³)	246	233	231	234	230	195	108	88	78	132	245	205
Ozone (µg/m³)	30	29	32	37	43	32	26	25	25	27	31	36
NO2 (µg/m³)	22	19	20	20	21	19	13	11	11	12	14	16
Benzene (µg/m³)	6	4	4	4	3	3	3	3	3	4	4	5

Table 9 shows that PM_{10}, NO_2, and benzene recorded their highest levels in January, while ozone peaked in May; all four pollutants exhibited their lowest concentrations in September.

Fig. 7. Annual trend of minimum and maximum concentrations of the top four selected pollutants.



According to this chart, the annual trends in the minimum and maximum concentrations of four selected pollutants show distinct seasonal variation across the year.

Table 10: Minimum

	1	2	3	4	5	6	7	8	9	10	11	12
PM10 (µg/m³)	71	21	71	10	34	40	9	0.07	7	17	64	32
Ozone (µg/m³)	6	9	10	16	8	9	8	8	5	7	8	5
NO2 (µg/m³)	8	10	14	5	5	7	1	2	1	4	4	5
Benzene (µg/m³)	2	2	1	2	1	1	0	0	1	1	1	1

Table 10 shows that all four pollutants recorded their lowest concentrations during the July–September period, with benzene reaching 0 µg/m³ in both July and August. An exceptionally low PM_{10} value of 0.07 µg/m³ was also observed. Such a value is likely influenced by measurement limitations, data recording issues, or sensor-related anomalies rather than representing actual ambient conditions. In contrast, a minimum PM_{10} concentration of 7 µg/m³ was recorded, which is considered plausible under favourable meteorological conditions, such as rainfall and enhanced atmospheric dispersion.

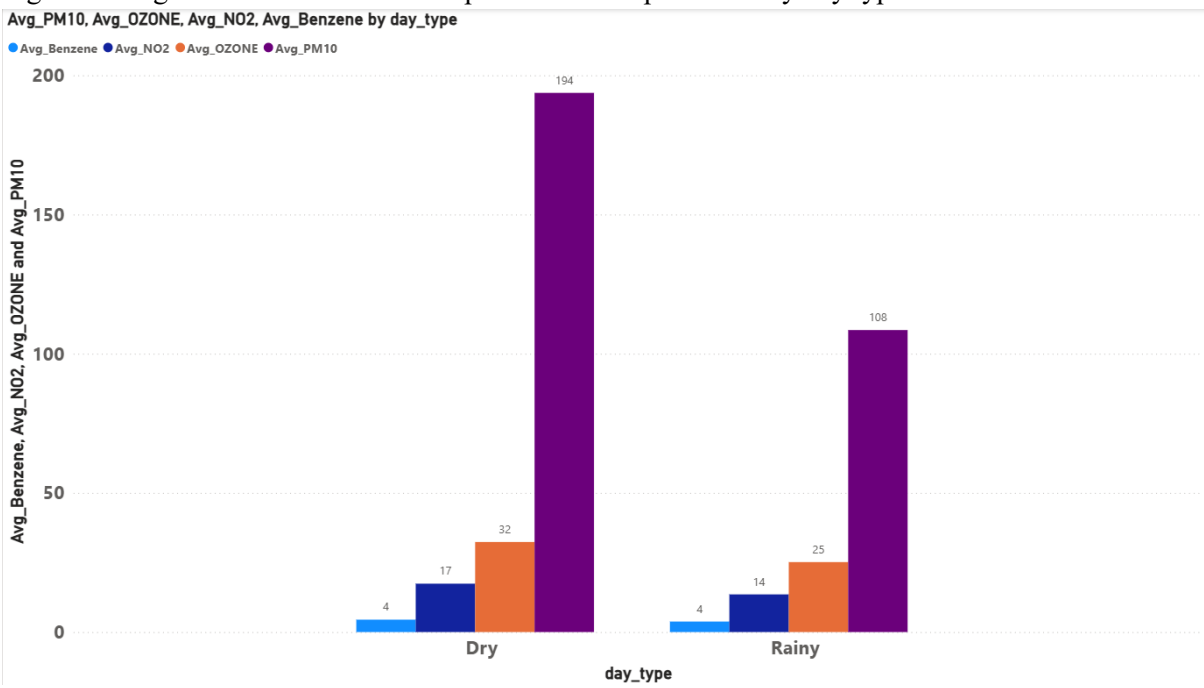
Table 11: Maximum

	1	2	3	4	5	6	7	8	9	10	11	12
PM10 (µg/m³)	792	743	534	674	688	773	542	326	274	549	893	585

Ozone (µg/m³)	71	62	62	85	104	88	87	93	78	72	110	116
NO2 (µg/m³)	72	32	30	38	64	54	30	23	23	31	56	39
Benzene (µg/m³)	17	9	7	7	9	12	7	8	9	24	12	15

Table 11 shows that PM_{10} recorded its highest concentration in November and its lowest in September. Ozone levels exceeded 60 µg/m³, NO_2 concentrations were above 22 µg/m³, and benzene levels were above 6 µg/m³.

Fig. 8. Average concentrations of the top four selected pollutants by day type in 2024.



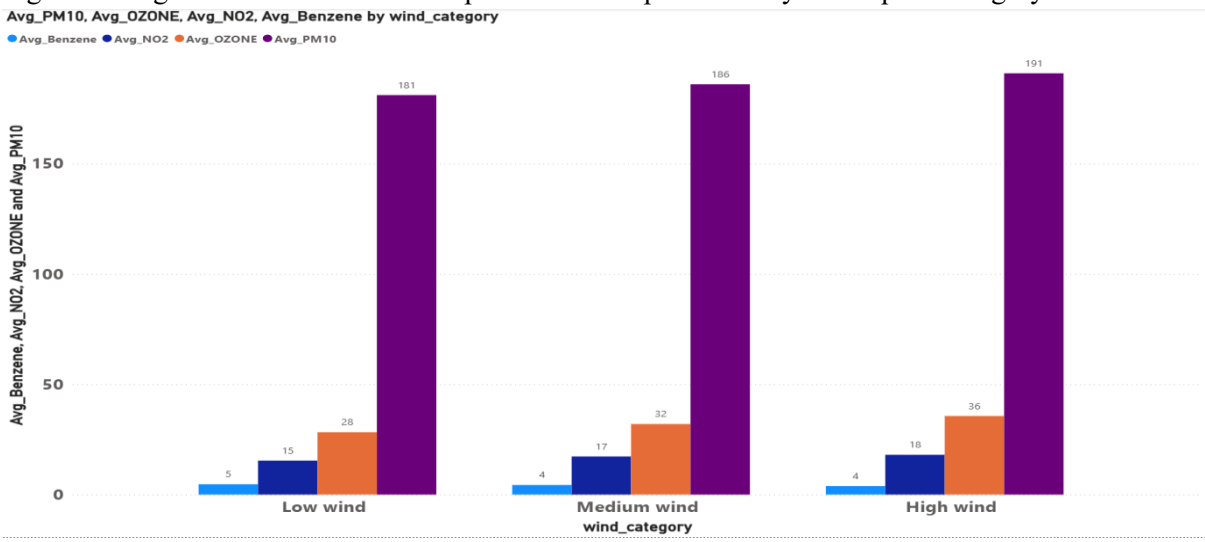
According to this chart, the average concentrations of the top four selected pollutants vary across different day types in 2024.

Table 12:

	Benzene (µg/m³)	NO2 (µg/m³)	Ozone (µg/m³)	PM10 (µg/m³)
Dry	4	17	32	194
Rainy	4	14	25	108

Table 12 shows that pollution levels were higher on dry days compared to rainy days.

Fig. 9. Average concentrations of the top four selected pollutants by wind speed category in 2024.



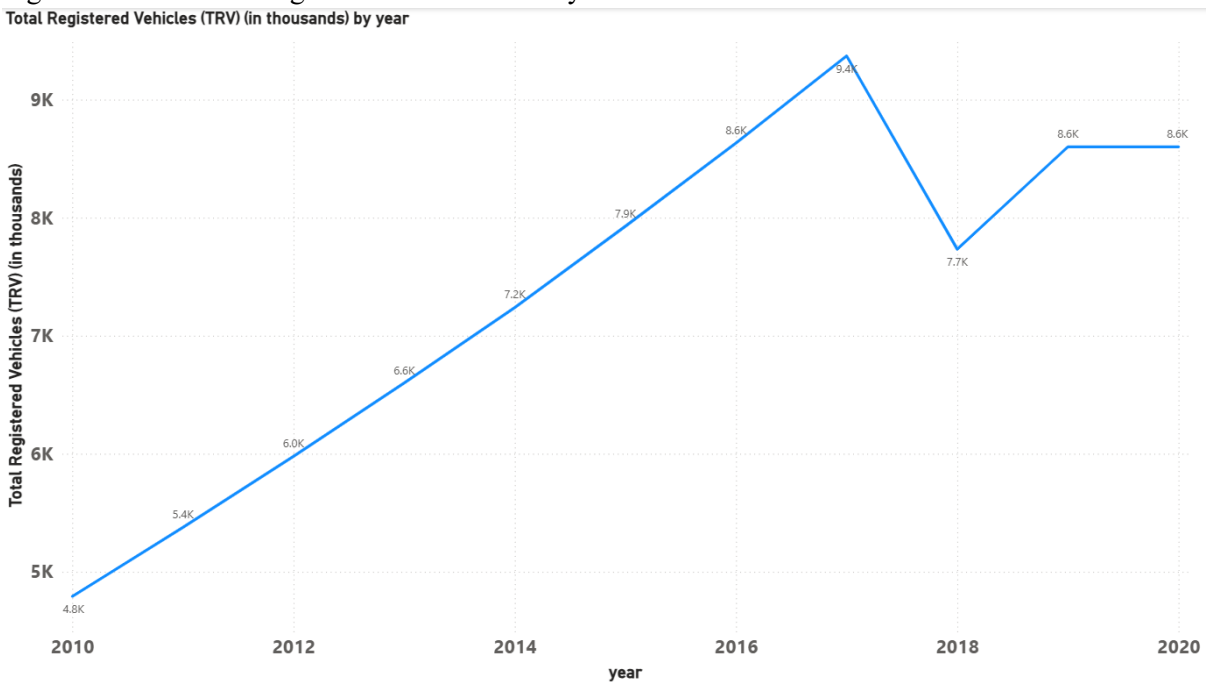
According to this chart, the average concentrations of the top four selected pollutants vary across different wind speed categories in 2024.

Table 13:

	Benzene (µg/m³)	NO2 (µg/m³)	Ozone (µg/m³)	PM10 (µg/m³)
Low wind (Km/h)	5	15	28	181
Medium wind (Km/h)	4	17	32	186
High wind (Km/h)	4	18	36	191

Table 13 shows that overall pollution levels increase with rising wind speed, whereas benzene concentrations decrease at higher wind speeds. This pattern for PM_{10} and NO_2 may be attributed to increased road dust resuspension and cross-regional transport of pollutants under stronger wind conditions, rather than indicating deteriorating local emission sources.

Fig. 10. Trend of total registered vehicles in Haryana from 2010 to 2020.



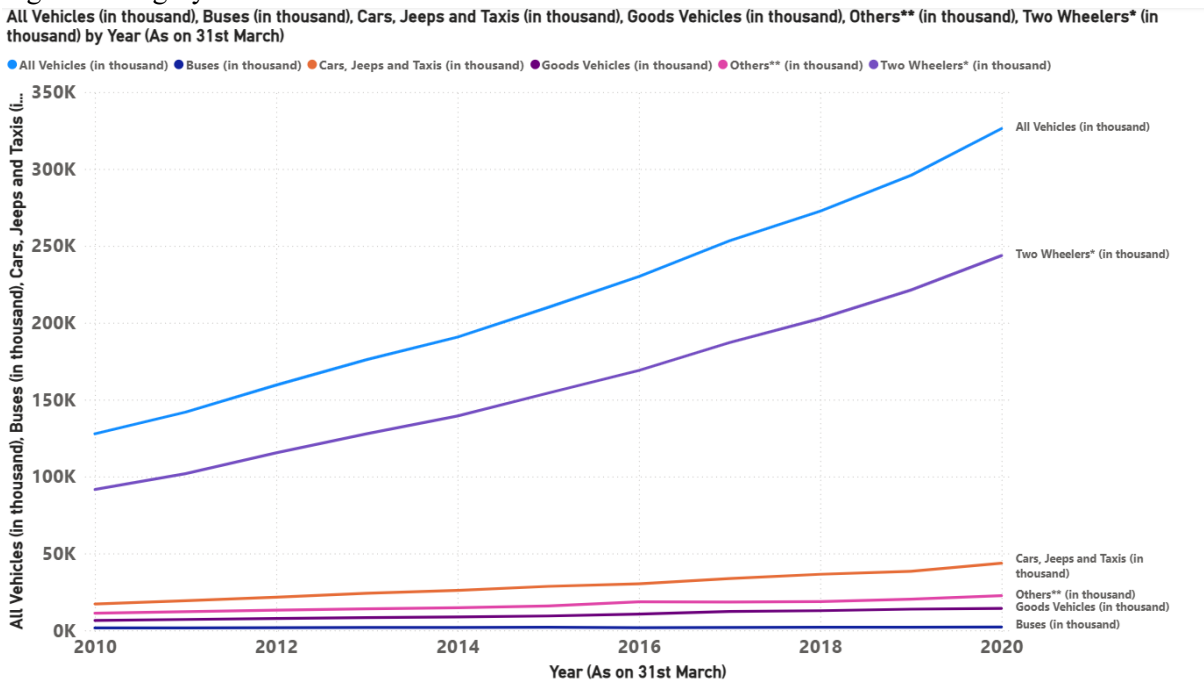
According to this chart, the total number of registered vehicles in Haryana shows a consistent upward trend from 2010 to 2020.

Table 14:

Year	Total Registered Vehicles (in thousand)
2010	4792
2011	5377
2012	5978
2013	6600
2014	7239
2015	7928
2016	8633
2017	9370
2018	7733
2019	8600
2020	8600

Table 14 shows a consistent increase in total registered vehicles in Haryana from 2010 to 2017, followed by a notable decline in 2018. This decline may reflect revised counting methodologies, de-registration of older vehicles, or data reporting changes in the MoRTH dataset rather than an actual reduction in the vehicle fleet. The identical values recorded for 2019 and 2020 further suggest a possible data reporting limitation, potentially related to administrative disruptions during the COVID-19 period, and should be interpreted with caution.

Fig. 11. Category-wise trend in the increase of vehicle counts in India from 2010 to 2020.



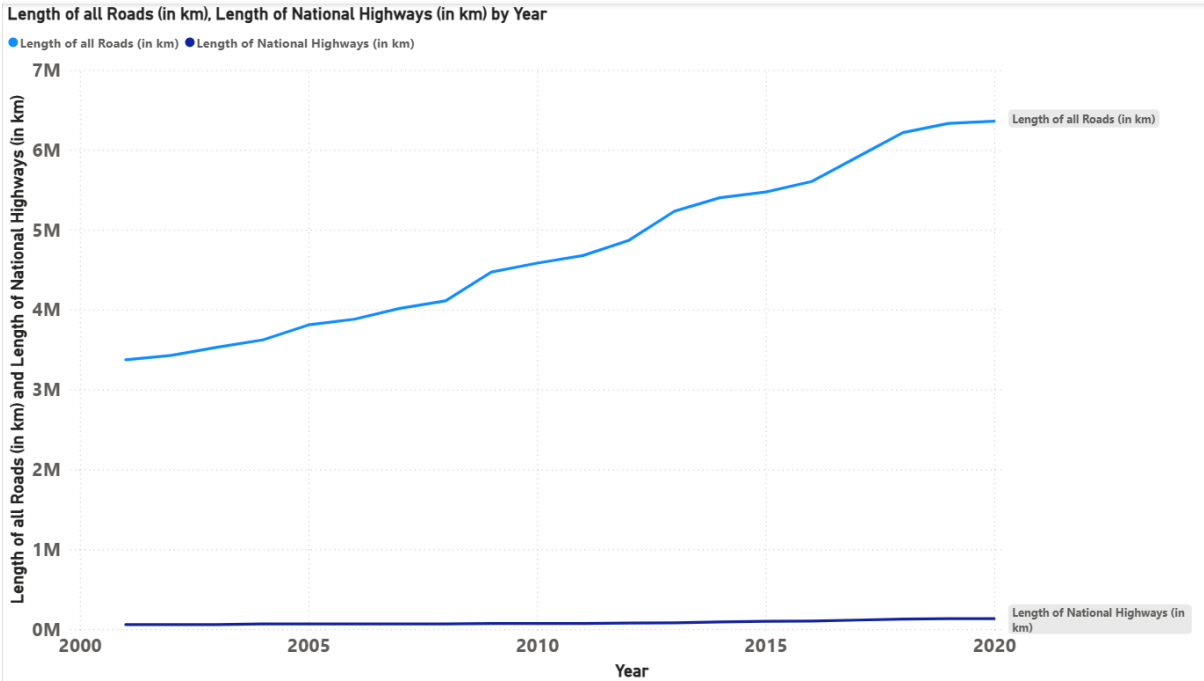
According to this chart, the category-wise vehicle counts in India show a clear increasing trend from 2010 to 2020.

Table 15: All values in Thousand

	All Vehicles	Two Wheelers	Cars, Jeeps, Taxis	Other Vehicles	Good Vehicles	Buses
2010	127746	91598	17109	11080	6432	1527
2011	141866	101865	19231	12102	7064	1604
2012	159491	115419	21568	13169	7658	1677
2013	176044	127830	24056	14037	8307	1814
2014	190704	139410	25998	14712	8698	1887
2015	210023	154298	28611	15799	9344	1971
2016	230031	168975	30242	18541	10516	1757
2017	253311	187091	33688	18411	12256	1864
2018	272587	202755	36453	18663	12773	1943
2019	295772	221270	38433	20254	13766	2049
2020	326299	243682	43650	22483	14288	2196

Table 15 shows a continuous increase in vehicle numbers over the years, indicating the dominance of private vehicle ownership compared to other vehicle categories. In contrast, buses exhibit the lowest growth rate, with a compound annual growth rate (CAGR) of 4% over the study period.

Fig. 12. Length of all roads and national highways (in km) from 2001 to 2020.



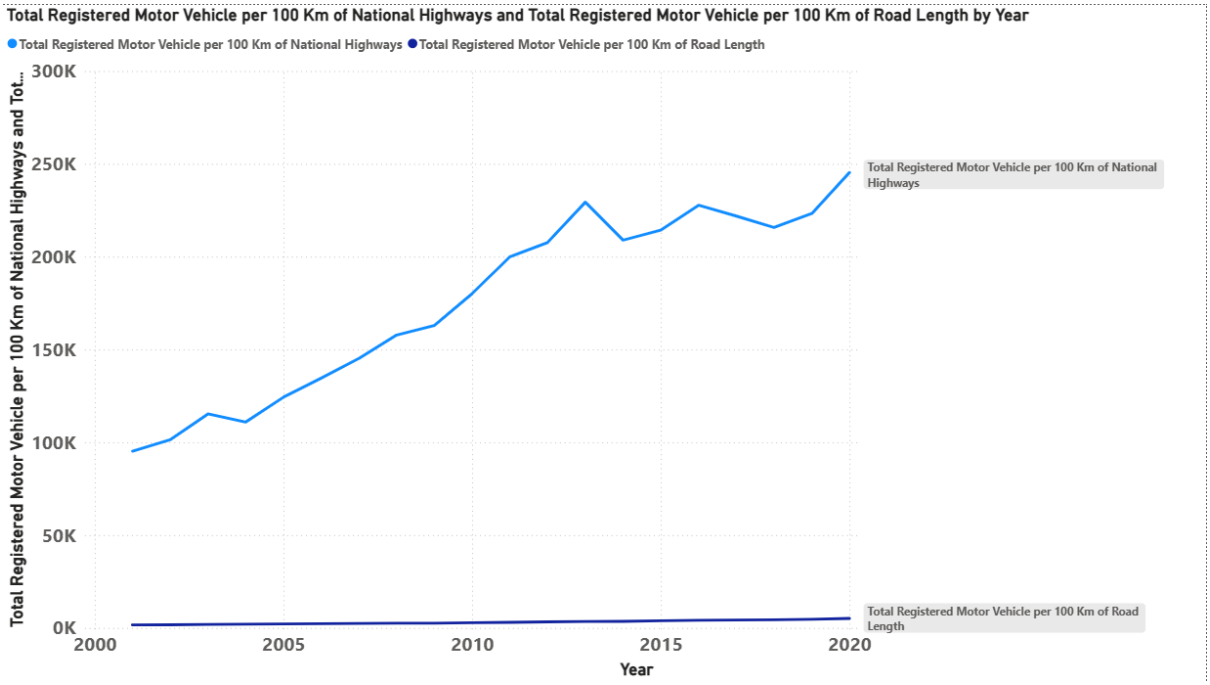
According to this chart, the length of all roads and national highways shows a consistent increasing trend from 2001 to 2020.

Table 17:

Year	Length of National Highways (in km)	Length of all Roads (in km)
2001	57737	3373520
2002	58112	3426600
2003	58112	3528654
2004	65569	3621507
2005	65569	3809156
2006	66590	3880651
2007	66590	4016401
2008	66754	4109592
2009	70548	4471510
2010	70934	4582439
2011	70934	4676838
2012	76818	4865394
2013	79116	5231922
2014	91287	5402486
2015	97991	5472144
2016	101011	5603293
2017	114158	5909561
2018	126350	6215797
2019	132500	6331757
2020	132995	6360004

Table 16 shows an approximate growth of 88.5% in the total length of all roads and about 130% growth in the length of national highways from 2001 to 2020, indicating that national highway expansion outpaced overall road network growth during the study period.

Fig. 13. Total registered vehicles per 100 km of all road length and national highways from 2001 to 2020.



According to this chart, the total number of registered vehicles per 100 km of all roads and national highways shows a steady increasing trend from 2001 to 2020.

Table 17:

Year	Total Registered Motor Vehicle per 100 Km of Road Length	Total Registered Motor Vehicle per 100 Km of National Highways
2001	1630	95244
2002	1720	101397
2003	1899	115307
2004	2008	110903
2005	2140	124295
2006	2309	134582
2007	2408	145228
2008	2564	157823
2009	2571	162940
2010	2788	180091
2011	3033	199997
2012	3278	207621
2013	3469	229420
2014	3530	208906
2015	3838	214329

2016	4105	227728
2017	4286	221895
2018	4385	215740
2019	4671	223224
2020	5131	245347

Table 17 shows that the number of vehicles per 100 km of national highways increased by approximately 158%, while the corresponding growth for all roads was about 215% from 2001 to 2020.

Discussion

This section interprets the key findings of the study in relation to existing evidence, focusing on the mechanisms linking air quality deterioration, traffic growth, and private vehicle dependence in Gurugram and similar rapidly urbanising cities. Rather than replicating the descriptive findings presented in the Results, this discussion emphasises causal interpretation, comparison with global evidence, and policy implications.

The average PM_{10} concentration of 186 µg/m³ recorded in Gurugram during 2024 is approximately four times the WHO annual guideline of 45 µg/m³, placing it firmly within the "very poor" air quality category. This figure reflects a chronic, city-wide particulate burden rather than isolated pollution events. The extreme maximum of 893 µg/m³ represents more than 20 times the WHO guideline, confirming that residents are periodically exposed to hazardous short-term concentrations. These values are consistent with patterns reported for Delhi and other NCR cities during winter months, further validating Gurugram's status as a high-burden urban air quality zone. Meteorological conditions—particularly the moderate average wind speed (9 km/h at 10 m) and relative humidity of 63%—create conditions that can either trap or disperse pollutants depending on the season, as discussed further below. The anomalously low minimum values (including 0.1 µg/m³ for PM_{10} and zero values for CO, benzene, and NO_x) reflect data recording limitations rather than actual ambient conditions and have been treated accordingly in the analysis.

Seasonal analysis (Fig. 4 and Fig. 5) confirms that pollution levels peak during the first and fourth quarters, consistent with winter boundary layer compression and reduced atmospheric mixing that trap pollutants near the surface. Monsoon rainfall during the third quarter (Fig. 6) provides effective wet deposition, resulting in substantially lower PM_{10} concentrations. The consistently higher pollutant levels on dry versus rainy days (Fig. 8) reinforce this pattern and highlight the significance of meteorological drivers in determining ambient air quality. The relationship between wind speed and pollutants (Fig. 9) reveals nuanced dynamics: while benzene decreases with higher wind speed due to enhanced dispersion, PM_{10} and NO_2 show a counterintuitive increase. For PM_{10}, this is consistent with road dust resuspension under stronger winds—a phenomenon well-documented in dust-laden urban environments such as Delhi-NCR. For NO_2, the pattern may reflect advection of pollutants from surrounding industrial zones under higher wind conditions, though further source apportionment analysis is needed to confirm this interpretation.

Transport-related trends contextualise these pollution patterns within a broader structural challenge. Vehicle registrations in Haryana grew at an estimated CAGR of approximately 6% between 2010 and 2017 (prior to the anomalous 2018 data point discussed in the Results). At the national level, two-wheelers and cars expanded at a CAGR of approximately 10% annually, while buses grew at only 4%, reinforcing deep structural underinvestment in public transportation (Fig. 11, Table 16). This pattern directly increases the emission burden per unit of transport demand. Road infrastructure expansion

(Fig. 12 and Fig. 13) has not kept pace with vehicle growth: while total road length grew by 88.5% and national highway length by 130% between 2001 and 2020, the number of registered vehicles per 100 km of roads increased by 215%, and by 158% for national highways alone. This persistent supply-demand imbalance indicates that new road construction alone cannot resolve congestion without concurrent demand-side management. The predominantly PM_{10}-driven pollution profile of Gurugram, combined with the dominance of high-growth private vehicle categories, strongly implicates non-exhaust emissions—including road dust resuspension and tyre wear particles—as significant contributors that will persist even under widespread vehicle electrification if fleet mass and vehicle density continue to rise.

Collectively, these findings indicate that rising pollution levels are closely linked to rapid growth in private vehicle ownership, insufficient public transport expansion, and increasing traffic density. These trends have significant implications for public health and environmental sustainability, underscoring the need for integrated traffic management strategies, improved public transportation systems, and evidence-based urban planning policies (MoRTH, 2023).

Evidence from World Bank–supported pollution control programmes demonstrates that sustained reductions in particulate matter can yield substantial public health benefits. For example, long-term air quality interventions in Mexico City resulted in a reduction in particulate pollution of over 70%, accompanied by measurable declines in respiratory and cardiovascular diseases and premature mortality (World Bank, 2023). Such findings reinforce the potential health gains associated with effective pollution control policies in highly urbanised settings.

According to the World Health Organization, approximately 99% of the global population is exposed to air quality levels exceeding recommended guidelines, contributing to an estimated 7 million premature deaths annually worldwide (WHO, 2023b). The burden of air pollution is disproportionately borne by low- and middle-income countries, which account for nearly 89% of pollution-related premature deaths. These impacts highlight the importance of systemic interventions, as many major pollution sources lie beyond individual control (CDC, 2023; WHO, 2023b; World Bank, 2023).

In response to traffic-related pollution, several international cities have implemented targeted policy interventions. Cities such as Singapore, London, and Stockholm have achieved measurable improvements in air quality through congestion pricing mechanisms, resulting in traffic reductions of 13–25%, travel delay reductions of approximately 14–30%, and reductions in exhaust emissions of around 14% within urban areas (SFCTA, 2020; FHWA, 2019). However, previous research suggests that congestion pricing alone may not constitute a permanent long-term solution due to rebound effects, whereby suppressed or new trips re-emerge over time (Metz, 2018). Consequently, such measures are most effective when combined with complementary policies that limit private vehicle dependence and strengthen alternative transport modes.

Beyond pricing mechanisms, non-pricing interventions—including expansion of public transportation, stricter vehicle emission standards, promotion of electric and low-emission vehicles, development of non-motorised transport infrastructure, traffic demand management, and public awareness initiatives—have demonstrated effectiveness in reducing traffic-related pollution in several international contexts (CLC, 2023; LTA, 2023; Transportstyrelsen, 2023a, 2023b; Local Government Association, 2024). Together, these integrated approaches illustrate that multidimensional transport and environmental policies are essential for achieving sustained reductions in urban air pollution and improving public health outcomes.

Several limitations of this study should be acknowledged. First, the study relies entirely on secondary datasets, as primary and ward-level pollution and health data for Gurugram were not publicly available. This constrains the spatial resolution of the analysis and prevents neighbourhood-scale or health-impact modelling. Second, the air quality data cover a single calendar year (2024), which limits the ability to assess long-term trends in Gurugram's pollution profile. Third, Gurugram-specific vehicle registration data are unavailable; Haryana state and India national data are therefore used as proxies, which may not fully capture city-level transport dynamics. Fourth, the analysis is descriptive and does not employ formal statistical modelling or causal inference techniques to establish relationships between vehicle growth and air quality outcomes. Fifth, the anomalies in the Kaggle dataset (zero values, 0.07 $\mu g/m^3$ PM_{10}) and in the MoRTH data (2018 dip, identical 2019–2020 values) introduce uncertainties that should be addressed through primary data collection or direct engagement with monitoring agencies in future research. Despite these constraints, the findings provide meaningful and policy-relevant insights into city-level pollution dynamics and motorisation trends applicable to Gurugram and comparable rapidly urbanising Indian cities.

Conclusion

The findings of this study identify PM_{10} as the dominant air pollutant in Gurugram, with concentrations increasing markedly during the winter season and declining during the monsoon period, largely due to rainfall-induced removal of particulate matter. Variations in pollutant levels with changing wind speeds further indicate complex interactions between meteorological conditions and pollutant dispersion. In parallel, the analysis reveals a sustained increase in total registered vehicles across India and Haryana, along with a substantially higher expansion of national highways compared to the overall road network.

The results also demonstrate a pronounced growth in private vehicle ownership, particularly two-wheelers and cars, while public transport vehicles and buses exhibit the lowest average annual growth. This imbalance between private vehicle expansion and public transport development has contributed to increasing traffic congestion, deteriorating air quality, and heightened risks to environmental and public health. These impacts are especially pronounced in rapidly urbanising cities such as Gurugram, underscoring the need for integrated urban planning, strengthened public transportation systems, and coordinated environmental and transport policies.

The study further indicates that non-pricing interventions—such as improvements in public transport infrastructure, traffic demand management strategies, and behavioural change initiatives—are likely to provide more sustainable long-term benefits than congestion pricing alone. In addition, effective reduction of PM_{10} concentrations and exposure to harmful compounds associated with tyre wear particles (TWPs) requires advancements in tyre material design, promotion of lighter vehicle technologies, and the development of regulatory frameworks addressing non-exhaust emissions.

Finally, future research incorporating primary data collection and high-resolution, locality-level datasets would enable more detailed assessments of neighbourhood-scale pollution patterns and associated health impacts. Such efforts would support more targeted, evidence-based urban and environmental interventions aimed at improving air quality and public health outcomes.

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